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Morphological and Biochemical Attributes of *Brassica cretica* Populations Grown Under Drought Tolerance Conditions

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Abstract

Drought stress is a major constraint on crop productivity in the Mediterranean region. *Brassica* crops are particularly valued in this region for their adaptability, nutritional benefits, and economic importance in sustainable farming systems. However, their productivity is highly sensitive to water deficits, necessitating the identification of drought-resilient genotypes. This study investigated the responses of five wild *Brassica cretica* populations and a commercial *Brassica oleracea* cultivar to a 50% reduction in irrigation, evaluating key physiological traits, leaf nutrient composition, and antioxidant activity. The experiment was conducted in the greenhouse facilities of the Laboratory of Vegetable Production, Agricultural University of Athens. The results revealed significant variation in drought tolerance among the tested populations. Specifically, an ecotype of *B. cretica* subsp. *cretica* (C: Akrokorinthos) and *B. cretica* subsp. *laconica* (E) showed substantial reductions in biomass, leaf area, and leaf number, whereas *B. cretica* subsp. *aegaea* (A: Manikia and B: Ymittos) and another ecotype of *B. cretica* subsp. *cretica* (D: Lasithi) maintained stable growth under water-limited conditions. Water deficit also significantly impacted leaf mineral composition, increasing NO_3^- and Na^+ levels while decreasing P, Zn, and Mn. Additionally, drought stress enhanced antioxidant capacity and secondary metabolite production, as indicated by elevated ferric reducing antioxidant power, Trolox equivalent antioxidant capacity, total phenolic content, and total flavonoid content. Notably, the two studied populations of *B. cretica* subsp. *aegaea* (A: Manikia, B: Ymittos) and the population of *B. cretica* subsp. *cretica* from Lasithi (Crete) (D) exhibit promising drought tolerance, suggesting their potential for cultivation or breeding in water-limited environments. This research contributes to the broader effort of identifying favorable traits in crop wild relatives and to utilize these valuable genetic resources to develop climate-resilient crops for Mediterranean agriculture, where sustainable water use is critical for food security.

Keywords: abiotic stress tolerance; biomass; nutrient concentration; soilless culture; osmotic stress; antioxidant response; *Brassica* species; genotypes

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1. Introduction

The Mediterranean Basin is characterized by a semi-arid climate with highly seasonal precipitation, resulting in naturally limited water availability [1]. Water scarcity is a critical challenge in the Mediterranean region, driven by increasing anthropogenic pressure on water resources and the intensifying impacts of climate change. Rising agricultural, urban, and tourism-related water demands have further strained already limited freshwater supplies [2]. At the same time, climate change has intensified drought conditions, with rising temperatures and declining rainfall leading to more frequent and severe drought events [3]. Projections indicate a 10–40% drop in annual precipitation by the end of the century, accompanied by higher summer temperatures, worsening water deficits [4,5]. Meteorological drought poses a significant threat to Mediterranean agriculture [6] and is considered one of the most damaging abiotic stresses, with the potential to reduce average crop yields by more than 50% [7,8]. In the Mediterranean area, where smallholder farms dominate [9], the socioeconomic impacts are especially severe, threatening food security and rural economies. One of the strategies that farmers should adopt to increase agricultural yields and farm returns, positively contributing to food security in the area, is the use of drought-resistant crops [10].

Plants exhibit a range of morphological, physiological, and biochemical responses to adapt to and survive under stress conditions. In particular, water scarcity impairs photosynthetic performance by restricting CO_2 diffusion into drought-affected leaves, mainly due to reduced conductance in both stomatal and mesophyll tissues [11]. Indeed, under drought stress, stomatal closure reduces transpiration to conserve water, but this also limits CO_2 uptake, decreasing carbon fixation efficiency, ultimately reducing crop yield [12]. Anatomical plasticity in both leaves and roots plays a critical role in enhancing crop resilience to drought stress. It has been documented that increased leaf cuticle thickness and greater epidermal layer thickness are key anatomical adaptations associated with drought tolerance [13]. Nutrient uptake is also compromised under water deficit, as low soil moisture induces stomatal closure, reduces transpiration, and consequently limits water and nutrient transport by slowing nutrient diffusion toward the root surface [14]. Under more severe drought, this restriction intensifies, further reducing nutrient availability to the root system [15]. Interestingly, drought stress in leafy vegetables has been linked to increased concentrations of vitamins, phenolic compounds, other bioactive metabolites, and enhanced antioxidant activity [16]. Although these changes may improve nutritional quality, they are often accompanied by reduced biomass production, highlighting a trade-off between stress tolerance and yield [17].

Brassica oleracea and its wild relatives form the monophyletic *B. sect. Brassica* within the otherwise polyphyletic genus *Brassica* [18]. This group includes economically important crops such as broccoli, cauliflower, kale, and cabbage, which are staple vegetables in Mediterranean diets. *Brassica* species are well-adapted to the region's climatic conditions, but their productivity is increasingly threatened by water scarcity and soil salinity [19]. It has been documented that wild relatives possess valuable agronomic traits, including tolerance to cold and salinity [20], as many cultivated varieties are particularly sensitive to climate change induced stresses [21] and have been bred primarily for yield rather than stress tolerance. Moreover, a few wild *Brassica* species have been investigated to identify alleles conferring resistance mainly against biotic stresses [22]. Beyond their agronomic importance, *Brassica* vegetables are nutritionally significant, rich in glucosinolates, phenolic compounds, vitamins, and antioxidants also possessing anticancer properties [23,24]. *B. cretica* holds significant importance as the closest wild relative of cultivated *Brassica oleracea* [18,25–27]. It is distributed across the Aegean Islands and adjacent mainland regions of Greece and Turkey, exhibiting high variation in traits that complicate its classification.

Its high genetic and phenotypic diversity, shaped by geographic isolation and pollinator-driven selection, makes it a valuable genetic resource for breeding programs aimed at improving crop resilience. Additionally, some populations may represent feral forms of cultivated plants, highlighting its dynamic evolutionary role [26]. However, despite progress in understanding drought responses in cultivated *Brassica* species, almost no data exist for their wild relatives (including *B. cretica*), leaving a critical gap in knowledge regarding their potential stress-adaptation traits and limiting the effective use of wild germplasm for improving drought resilience.

B. cretica grows in crevices of limestone cliffs from sea level up to 1100 m a.s.l. It is a suffrutescent species, woody only at the base, reaching 50–150 cm in height. The leaves are petiolate to subsessile, simple or lyrate-pinnate, shallowly dentate, glabrous, and ±coriaceous. The inflorescence is a simple or sometimes branched raceme. Petals are 15–25 mm long and may be white, cream, or yellow. The fruits are erecto-patent siliquae, 4–10 cm long, beaked, containing numerous seeds arranged in a single row. Seeds are globose, 1.6–2.8 mm in diameter. Snogerup et al. [28] carried out a detailed study of variation in *B. cretica* and recognized three subspecies, which are rather well defined, although fully interfertile and sometimes forming intermediates in contact zones. Subsp. *aegaea* Snogerup, M.A. Gust. & Bothmer is characterized by all leaves being short-petiolate to sessile, petals usually yellow, and seeds only faintly reticulate. Subsp. *cretica* has lower leaves long-petiolate, basally pinnatisect or undivided, petals white or rarely cream, and seeds prominently reticulate. Subsp. laconica M.A. Gust. & Snogerup differs by having most lower leaves short-petiolate, pinnatifid to pinnatisect, petals pale yellow or cream, and seeds prominently reticulate. All three subspecies have been assessed under the IUCN criteria and were evaluated as Least Concern (LC) [29–31]. *B. cretica* is the closest wild relative of cultivated *B. oleracea*, thus it is proposed to be the ancestor of many types of commercially important vegetables, including broccoli, Brussels sprouts, cabbage, cauliflower, kale, swede, turnip and oilseed rape [26]. *B. cretica* can provide valuable agronomic traits and act as a reservoir of useful alleles that may have disappeared during domestication [22]. However, Stathi et al. [27] reported that only a limited number of *B. cretica* accessions are currently preserved in Greek gene banks. Hence, prioritizing ex situ conservation efforts, such as seed collection and vegetative propagation of Greek populations, is essential.

Considering the intensifying drought conditions across the Mediterranean region and the agronomic and nutritional significance of *Brassica* crops, this study evaluated the effects of a 50% water deficit on growth, yield, nutrient composition, and antioxidant capacity in selected *B. cretica* subspecies populations alongside a commercial *B. oleracea* cultivar, with the aim of identifying populations exhibiting superior drought tolerance. To our knowledge, this is the first report examining how drought stress alters the morphological, physiological, and biochemical characteristics of these *B. cretica* subspecies. The study addressed two main research questions: (i) how wild and cultivated populations differ in their responses to drought, and (ii) which traits best distinguish drought-tolerant from drought-sensitive populations. We hypothesized that tolerant genotypes would maintain higher biomass and leaf area under stress and exhibit improved nutrient retention along with elevated antioxidant and phenolic concentrations compared with sensitive genotypes.

2. Materials and Methods

2.1. Experimental Design

The experiment was conducted in the greenhouse facilities of the Laboratory of Vegetable Production at the Agricultural University of Athens (AUA), located at 37°59′2″ N latitude and 23°42′19″ E longitude. The study involved six populations: five wild popula-

tions of *B. cretica*, including two *B. cretica* subsp. *aegaea* populations from Evia and Ymittos, two *B. cretica* subsp. *cretica* populations from Akrokorinthos and Crete, one *B. cretica* subsp. *laconica* population from Leonidio, and one commercial variety of *B. oleracea* var. *rubra* cv. 'Vertus 2' (Garden Expert S.A., Koropi, Greece) (Table 1) [27]. The seeds of *wild* populations were collected from their natural habitats. Plants were cultivated in an open soilless system under two different irrigation regimes (100% serving as control and 50% serving as drought stress, respectively).

Table 1. Sampling information for *B. cretica* accessions, including taxonomic identity, collection locations and dates.

Code	Taxon	Collection Location	Sampling Date	Latitude	Longitude
A	*B. cretica* subsp. *aegaea*	Manika, Evia, Greece	June 2022	38°32.582′	24°1.052′
B	*B. cretica* subsp. *aegaea*	Ymitos, Attica, Greece	June 2022	37°56.671′	23°48.030′
C	*B. cretica* subsp. *cretica* or *nivea*	Akrokorinthos, Corinthia, Greece	May 2022	37°53.363′	22°52.138′
D	*B. cretica* subsp. *cretica*	Lasithi, Crete, Greece	May 2021	35°19.772′	25°41.458′
E	*B. cretica* subsp. *laconica*	Leonidio, Arcadia, Greece	June 2022	37°9.940′	22°52.255′
F	*Brassica oleracea* var. *rubra* cv. 'Vertus 2'	Commercial seed source, EU	Not specified		

2.2. Seedling Propagation, Cultivation Conditions and Nutrient Solution Composition

Seedling production was carried out at the Laboratory of Vegetable Production, Agricultural University of Athens. On 7 October 2022, seeds from *B. cretica* subspecies and the commercial variety were sown individually into AO 25/40 rockwool plugs (Rockwool B.V., Roermond, The Netherlands), followed by the application of a thin vermiculite overlay. On 28 October 2022, seedlings at the two- to three-leaf stage were transplanted into 33 L perlite bags to establish an open hydroponic system. Twenty-four hours before transplantation, the perlite bags were pre-saturated with a starter solution (Table 2). After saturation, drainage holes were made at the base of each bag to facilitate excess nutrient solution removal. Throughout the growing season, the mean day and night temperatures were maintained at 18–22 °C and 12–16 °C, respectively, while relative humidity ranged between 60 and 75%. The plants were cultivated under natural daylight conditions, without the use of supplemental lighting.

Throughout the experiment, the rockwool plugs were manually irrigated every 1–2 days with a nutrient solution optimized for vegetative development. The solution was formulated using NUTRISENSE (https://nutrisense.online/, accessed on 7 October 2022), a specialized nutrient management software. Stock solutions containing the requisite macro- and micronutrients were initially prepared at elevated concentrations and subsequently diluted at a 1:100 ratio prior to application. The nutrient formulation utilized in this study was standardized for all plants, whereas irrigation volumes differed between treatments. The concentrations of specific nutrients utilized at various growth stages are presented in Table 2.

Table 2. Chemical composition of the nutrient solutions utilized during the experiment.

Nutrient Solution Composition	Starter Solution (26 October 2022)	Vegetative Phase (28 October 2022)	Unit
EC	3.00	2.40	dS/m
pH	5.60	5.60	
NO_3^-	15.05	14.00	mM
K^+	8.72	7.00	mM
Ca^{2+}	5.25	5.00	mM
Mg^{2+}	3.42	2.00	mM
SO_4^{2-}	5.02	3.06	mM
$H_2PO_4^-$	1.65	1.50	mM
NH_4^+	0.89	0.82	mM
Fe	40.00	30.00	µM
Mn^{2+}	5.00	10.00	µM
Zn^{2+}	7.00	7.00	µM
B	40.00	35.00	µM
Cu^{2+}	0.80	0.80	µM
Mo	0.70	0.80	µM
Cl^-	0.40	0.40	µM
K/(K + Ca + Mg)	0.50	0.50	mol/mol
Ca/(K + Ca + Mg)	0.30	0.36	mol/mol
Mg/(K + Ca + Mg)	0.20	0.14	mol/mol
N/K	1.83	2.12	mol/mol
NH_4-N/Total-N	0.06	0.06	mol/mol

Two irrigation regimes were initiated at transplanting: (1) a control group receiving 100% of water capacity, and (2) a deficit-irrigated group receiving 50% of the control volume. As the experiment was conducted under soilless cultivation, the control plants were irrigated with 130% of the required volume to allow for 30% drainage. Therefore, the deficit group received 50% of this total amount, equivalent to 65% of the calculated water requirement. Drought treatment was initiated one after transplanting, at the 2–3 leaf stage. The experiment included five replicates per treatment, with each replicate consisting of 5 perlite bags. In total, the experimental trial comprised 60 perlite bags (12 treatments × 5 replicates), with each bag accommodating five genetically uniform plants of the same ecotype. These perlite bags were placed on 10 hydroponic circuits, with each hydroponic circuit accommodating 6 perlite bags. Half (5) of the circuits received adequate water supply, while the other half were subjected to water limitations. Within each irrigation regime, the six ecotypes were randomly placed within each hydroponic circuit, so that each circuit contained one bag per ecotype. This arrangement corresponds to a randomized complete block design (RCBD), with the hydroponic circuits serving as blocks.

2.3. Growth and Yield Parameters

Plants were harvested 32 days after transplanting, once they reached marketable size. At harvest, all plants were collected, and leaf fresh weight was measured to determine

total yield. In addition to fresh biomass, the number of leaves per plant and total leaf area were recorded. Fresh weights were measured using a high-precision electronic scale (Mettler-Toledo, Columbus, OH, USA; accuracy: ±0.1 mg). For leaf area determination, leaves were manually separated and placed on the transparent conveyor belt of an LI-3100C area meter (LI-COR, Inc., Lincoln, NE, USA). After harvest, plant samples were dried in a laboratory oven at 65 °C until a constant dry weight was attained. Dry matter content was calculated as the difference between the fresh mass before drying and the mass loss after drying [32].

2.4. Nutrient Concentration in Plant Tissues

The dried shoot material was subsequently pulverized and subjected to dry ashing at 550 °C for 8 h. The resulting ash was dissolved in 0.1 N hydrochloric acid (HCl) for subsequent analysis. The concentrations of phosphorus (P) in the plant extract samples were determined colorimetrically using the method developed by Murphy and Riley [33]. Potassium (K) and sodium (Na) concentrations were quantified using a Sherwood 410 flame photometer (Sherwood, Cambridge, UK), while iron (Fe), zinc (Zn), copper (Cu), and manganese (Mn) concentrations were determined using an atomic absorption spectrophotometer (AA-7000, Shimadzu Co., Tokyo, Japan). For the determination of nitrates (NO_3), the dried plant tissues were subjected to hot water extraction, following the procedure previously described by Asch et al. [34]. The nitrate concentration was subsequently quantified through the salicylic acid method [35].

2.5. Determination of Plant Biochemical Profile

Fresh plant tissues were freeze-dried and then subjected to methanolic extraction following the procedure outlined by Karavidas et al. [36]. To evaluate the antioxidant capacity of *Brassica* extracts, two assays were used: the ferric reducing antioxidant power (FRAP) method [37] and the Trolox equivalent antioxidant capacity (TEAC) assay [38]. Results from both assays were expressed as Trolox equivalents per gram of dry biomass. The total phenolic content in the samples was determined using the Folin–Ciocalteu reagent method, as described by Jan et al. [39], and expressed in milligrams of gallic acid equivalents (mg GAE) per gram of dry biomass. To quantify total flavonoids, the aluminum chloride colorimetric assay was performed according to the method of Safafar et al. [40], with results reported in milligrams of quercetin equivalents per gram of dry biomass.

2.6. Statistical Analysis

In this study, one-way analysis of variance (ANOVA) was performed within each *Brassica* population to assess the effects of water stress on fresh weight, leaf number, leaf area and dry matter content. To assess variations in nutrient concentrations (P, K, Na, NO_3) and biochemical compounds (FRAP, TEAC, TPC, TFC) across different *Brassica* populations and irrigation treatments, a two-way analysis of variance (ANOVA) was employed. For statistical analyses, each replicate corresponded to the mean value of the measured parameters of five plants grown within the same perlite bag. All statistical procedures were executed using STATISTICA software (version 12.0, Windows). In order to assess data normality, the Shapiro–Wilk test was used. In instances where drought treatment exhibited a statistically significant effect on the measured variables, mean comparisons were conducted using the Duncan Multiple Range Test, with statistical significance determined at $p \leq 0.05$ level.

3. Results

As shown in Figure 1, the populations that showed a decrease in leaf fresh weight (which corresponds to crop yield) under reduced irrigation conditions were C and E, by

31% and 29%, respectively, along with the commercial variety F, by 14%. In contrast, the remaining populations showed no significant reduction in leaf fresh weight due to lower irrigation levels.

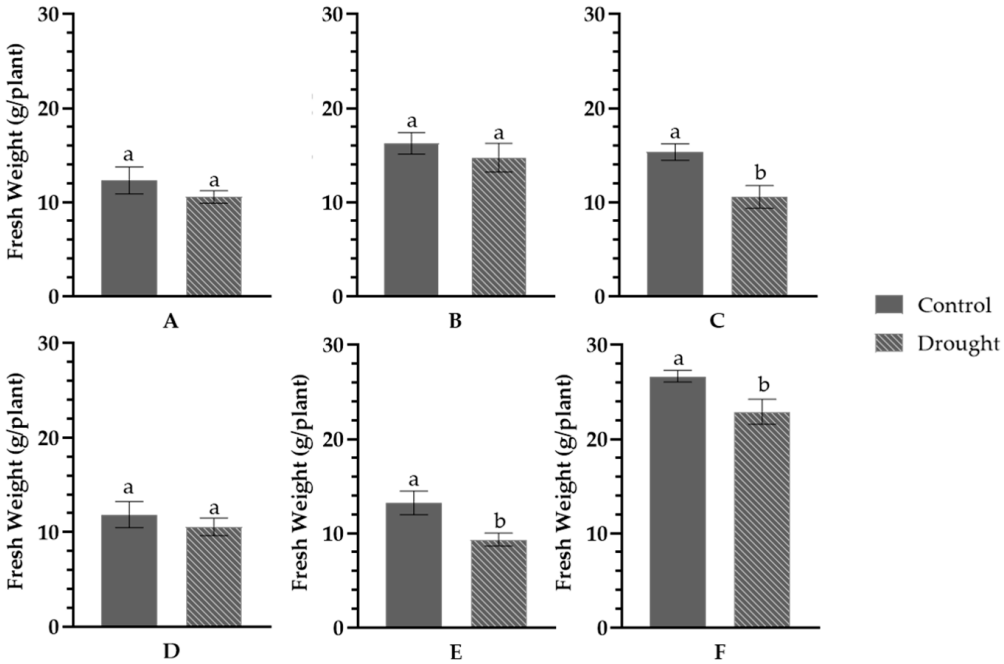

Figure 1. Impact of drought stress on the fresh weight per plant (g/plant) among various *Brassica* populations. The term 'Control' refers to optimal irrigation conditions, whereas 'Drought' indicates a 50% reduction in water supply. (**A**–**F**) indicate plant materials as listed in Table 1. Bars marked with different letters denote statistically significant differences according to Duncan's multiple range test ($p < 0.05$). Error bars represent the standard error of the means.

As demonstrated in Figures 2 and 3, the observed reduction in fresh weight in population C can be attributed to a decrease in both the number of leaves and leaf area. In contrast, the reduction observed in population E and variety F is attributed solely to a decreased leaf area under drought conditions. However, despite these reductions in growth-related parameters, populations C and E exhibited an increase in dry matter content, indicating a higher percentage of solid biomass in relation to water content (Table 3).

Table 3. Effect of reduced irrigation on dry matter content (DMC) of plants. A–F indicate plant materials as listed in Table 1.

		Populations					
Variable	**Irrigation Treatment**	**A**	**B**	**C**	**D**	**E**	**F**
DMC (%)	Control	7.98	8.83	8.89	10.21	8.95	8.83
	Drought	7.74	9.25	9.23	10.46	9.49	9.25
Statistical significance		NS	NS	*	NS	*	NS

Mean values within the same column that are followed by different letters differ significantly according to Duncan's Multiple Range Test at $p < 0.05$. * indicates levels of significance at $p < 0.05$, while 'NS' denotes a non-significant difference.

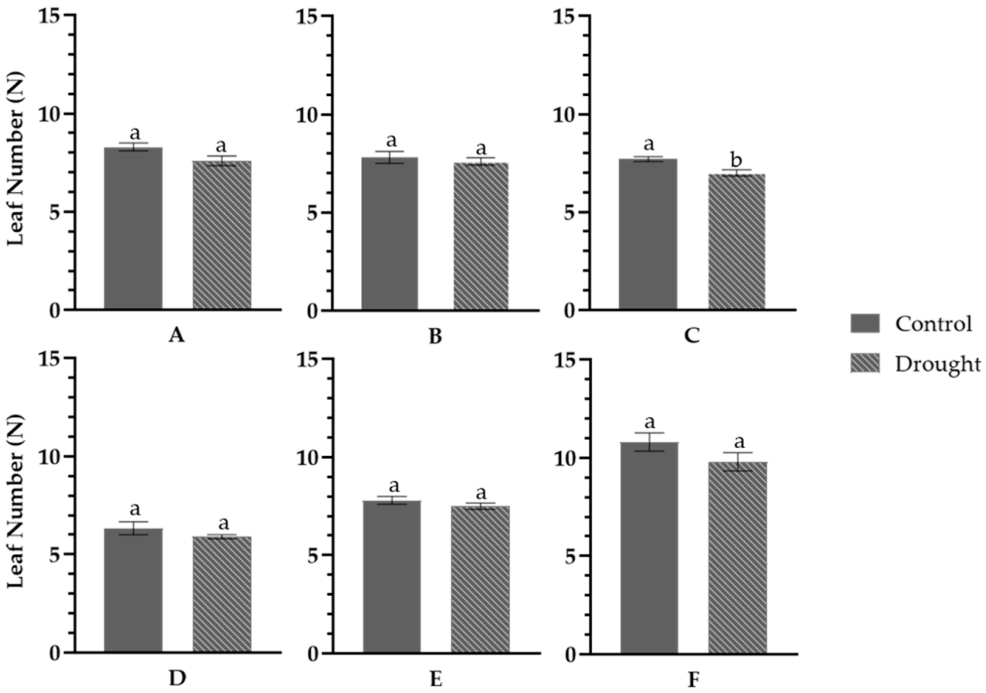

Figure 2. Impact of drought stress on Leaf Number (N) among various *Brassica* populations. The term 'Control' refers to optimal irrigation conditions, whereas 'Drought' indicates a 50% reduction in water supply. (**A**–**F**) indicate plant materials as listed in Table 1. Bars marked with different letters denote statistically significant differences according to Duncan's multiple range test ($p < 0.05$). Error bars represent the standard error of the means.

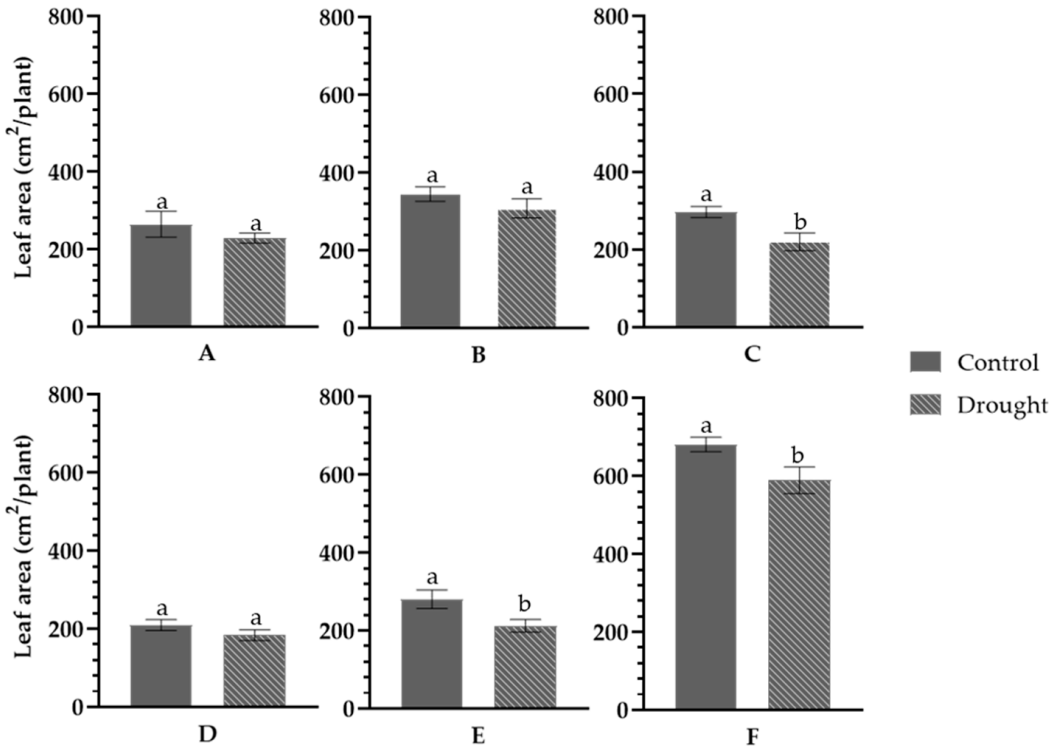

Figure 3. Impact of drought stress on Leaf area (cm^2/plant) among various *Brassica* populations. The term 'Control' refers to optimal irrigation conditions, whereas 'Drought' indicates a 50% reduction in water supply. (**A**–**F**) indicate plant materials as listed in Table 1. Bars marked with different letters denote statistically significant differences according to Duncan's multiple range test ($p < 0.05$). Error bars represent the standard error of the means.

As shown in Table 4, only the concentration of Na increased significantly in the shoot tissues of *Brassica* populations under drought conditions. Conversely, the concentrations of P and K exhibited no significant difference at 50% reduction in irrigation. Statistically significant differences were observed among the different populations for all macronutrients measured. It is noteworthy that populations D and E exhibited the highest P concentrations in their leaves. Population E exhibited the highest K concentration and the lowest Na accumulation in leaves. Notably, a significant interaction between irrigation level and population was observed only for P, with population E showing a significant reduction in leaf P concentration under drought stress. A substantial variation in NO_3 concentration was evident among the distinct populations within the fresh plant tissue. It is noteworthy that populations C and E, which also exhibited reduced growth, had the highest NO_3 concentrations, 1880 and 2015 mg/kg FW, respectively. Furthermore, the interaction between population and irrigation proved to be significant, with population C exhibiting a 28% increase in NO_3 concentration under drought stress.

Table 4. Effect of reduced irrigation on macronutrient concentrations (P, K, Na) (mg/g in dry weight) and NO_3 (mg/kg of fresh weight) in the shoots of *Brassica* populations evaluated. A–F indicate plant materials as listed in Table 1.

Treatment		P (%)	K (mg/g)	Na (mg/g)	NO_3 (mg/kg FW)
Main effects					
Irrigation	Population				
Control		0.62	41.75	1.68	1563
Drought		0.60	41.40	2.17	1640
	A	0.58 b	35.60 d	2.12 ab	1380 c
	B	0.59 b	38.00 cd	2.31 a	1423 c
	C	0.57 b	41.50 bc	1.91 bcd	1880 a
	D	0.68 a	42.80 b	1.67 cd	1666 b
	E	0.71 a	48.20 a	1.57 d	2015 a
	F	0.56 b	43.60 b	1.98 abc	1304 c
Interactions					
Control	A	0.60 c	35.40	0.85	1370 cd
	B	0.60 c	40.00	2.00	1442 bcd
	C	0.58 c	40.60	1.62	1647 bc
	D	0.66 b	39.00	1.48	1541 bc
	E	0.74 a	48.40	1.43	2039 a
	F	0.56 c	46.00	1.66	1367 cd
Drought	A	0.57 c	35.80	2.40	1390 cd
	B	0.57 c	36.00	2.62	1404 cd
	C	0.56 c	42.40	2.20	2112 a
	D	0.69 b	45.00	1.79	1715 b
	E	0.67 b	48.00	1.72	1992 a
	F	0.57 c	41.20	2.31	1230 d
Statistical Significance					
Irrigation		0.05	NS	***	NS
Populations		***	***	**	***
Irrigation × Populations		*	NS	NS	*

Mean values within the same column that are followed by different letters differ significantly according to Duncan's Multiple Range Test at $p < 0.05$. *, **, *** indicate levels of significance at $p < 0.05$, $p < 0.01$ and $p < 0.001$, respectively, while 'NS' denotes a non-significant difference.

As demonstrated in Table 5, under drought conditions, there was a significant decrease in the concentrations of the micronutrients Zn and Mn, with reductions of 20% and 14%, respectively. Statistically significant differences were observed among the different cultivated populations in the concentrations of all four micronutrients measured. Population A exhibited the highest concentrations of Fe, Zn and Mn, while the commercial variety demonstrated the highest concentrations in Cu. No statistically significant interaction was identified between the two factors (Irrigation x Populations).

Table 5. Effect of reduced irrigation on micronutrient concentrations (Fe, Zn, Mn, Cu) in the shoots of *Brassica* populations evaluated. A–E indicate plant materials as listed in Table 1.

Treatment		Fe (μg/g)	Zn (μg/g)	Mn (μg/g)	Cu (μg/g)
Main effects					
Irrigation	Populations				
Control		69.78	44.67	60.55	5.00
Drought		66.50	35.54	51.86	4.82
	A	71.19 a	45.96 a	64.35 a	4.33 b
	B	71.09 a	35.89 bc	63.01 ab	4.67 b
	C	67.20 ab	42.43 ab	57.73 bc	4.72 b
	D	61.42 b	41.72 ab	38.47 d	4.44 b
	E	68.13 ab	33.93 c	53.05 c	6.29 a
Statistical Significance					
Irrigation		NS	**	***	NS
Populations		*	***	***	***
Irrigation × Populations		NS	NS	NS	NS

Mean values within the same column that are followed by different letters differ significantly according to Duncan's Multiple Range Test at $p < 0.05$. *, **, *** indicate levels of significance at $p < 0.05$, $p < 0.01$ and $p < 0.001$, respectively, while 'NS' denotes a non-significant difference.

This study also assessed the antioxidant properties of methanolic *Brassica* extracts using two assays, Trolox Equivalent Antioxidant Capacity (TEAC) and Ferric Reducing Antioxidant Power (FRAP). The results of both tests indicated that plants grown under well-watered conditions exhibited the lowest antioxidant activity (Table 6). By contrast, reducing irrigation by 50% significantly enhanced the plants' antioxidant potential, with FRAP and TEAC values increasing by 23% and 4%, respectively. Furthermore, drought stress resulted in a 27% increase in total phenolic content and a 26% rise in total flavonoid content. Of the cultivated populations, the commercial variety F exhibited the highest values for all measured biochemical parameters, including FRAP, TEAC, total phenolic content (TPC) and total flavonoid content (TFC). The interaction between drought stress and population was significant for FRAP, TEAC and TPC. Specifically, in the FRAP assay, only population A showed no change in antioxidant capacity under drought conditions. Regarding TEAC, a significant increase was observed exclusively in the commercial variety F, while populations A and D did not exhibit an increase in TPC under drought stress.

Table 6. Effect of reduced irrigation on plant biochemical properties in the shoots of *Brassica* populations evaluated. A–F indicate plant materials as listed in Table 1. FRAP: Ferric Reducing Antioxidant Power; TEAC: Trolox Equivalent Antioxidant Capacity; TPC: Total Phenolic Content; TFC: Total Flavonoid Content.

Treatment		FRAP	TEAC	TPC	TFC
		Trolox (mg/g)		Gallic Acid (mg/g)	Quercetin (mg/g)
Main effects					
Irrigation	Populations				
Control		17.66	23.44	3.72	8.32
Drought		21.76	24.48	4.75	10.52
	A	15.31 e	21.68 d	3.03 e	8.69 bc
	B	15.76 e	21.04 d	3.39 d	8.28 c
	C	20.61 c	25.37 c	4.25 c	9.11 bc
	D	22.87 b	26.98 b	5.08 b	9.35 b
	E	19.21 d	21.55 d	4.31 c	8.86 bc
	F	27.21 a	28.41 a	6.06 a	12.96 a
Interactions					
Control	A	15.39 ef	21.86 de	2.82 d	7.90
	B	13.66 f	19.93 e	2.92 d	6.78
	C	18.05 d	24.76 c	3.55 c	7.84
	D	21.44 c	28.05 ab	4.80 b	8.36
	E	15.97 e	20.74 de	3.51 c	7.94
	F	22.93 bc	26.83 bc	5.20 b	11.23
Drought	A	15.24 ef	21.49 de	3.24 cd	9.48
	B	17.17 de	21.79 de	3.71 c	9.28
	C	22.66 bc	25.86 c	4.82 b	10.13
	D	23.94 b	26.18 bc	5.29 b	10.10
	E	22.44 bc	22.35 d	5.11 b	9.78
	F	30.63 a	29.68 a	6.74 a	14.35
Statistical Significance					
Irrigation		***	*	***	***
Populations		***	***	***	***
Irrigation × Populations		***	*	***	NS

Mean values within the same column that are followed by different letters differ significantly according to Duncan's Multiple Range Test at $p < 0.05$. *, *** indicate levels of significance at $p < 0.05$ and $p < 0.001$, respectively, while 'NS' denotes a non-significant difference.

4. Discussion

It is well established that drought stress exerts a deleterious effect on the growth and productivity of various *Brassica* species. In the current study, a 50% reduction in irrigation led to a noticeable decrease in fresh leaf weight in two of the five *B. cretica* populations (C and E), showing reductions of 31% and 29%, respectively, as well as in the commercial cultivar *B. oleracea* (F), which exhibited a 14% decline in leaf fresh weight. These findings are consistent with previous research indicating the sensitivity of brassicaceous crops to

drought conditions. For example, Akram et al. [41] reported that drought stress at 60% field capacity (FC) significantly reduced shoot, fresh, and dry biomass in both 'Parasol' and 'Grand Slem' broccoli cultivars. Similarly, Ibrahim et al. [42] observed that water-stressed broccoli plants displayed significantly lower vegetative growth and yield characteristics than their well-watered counterparts. Seymen et al. [43] also suggested that water deficiency can adversely affect cabbage growth. Such reductions can be attributed to limited water availability, which restricts cell division, elongation, and differentiation, ultimately constraining overall plant growth and development [44]. However, it is noteworthy that in this study, no significant changes in fresh weight were observed in *B. cretica* populations A, B, and D. Additionally, Yoo et al. [45] reported that among several *Brassica* species, namely *B. rapa*, *B. juncea*, *B. nigra*, *B. napus*, and *B. oleracea*, no reductions in fresh weight were noted under drought conditions, with the exception of *B. carinata*, which did exhibit a decline. *B. carinata*-derived lines (CDLs) developed in a *B. juncea* background carry genomic segments that enhance tolerance to moisture-deficit stress. Several CDL genotypes showed strong drought resilience and heterosis, by maintaining biomass production, a trait that is linked to improved water use efficiency and a robust root architecture that supports optimal absorption of water and nutrients [46]. In addition, Vennam et al. [47] reported that Ethiopian mustard (B. carinata) genotypes exhibiting higher root-to-shoot ratios generally develop deeper root systems, which enhance water acquisition and contribute to better stomatal control and lower leaf temperatures. The diversity of responses highlights the genetic variability in drought tolerance within and among *Brassica* species.

Drought stress typically causes a reduction in the number of leaves, leaf area, and both fresh and dry biomass in *Brassica* species, as observed in Chinese cabbage (*Brassica rapa* L. spp. *pekinensis*), where plants subjected to drought stress had significantly fewer leaves and reduced leaf area compared to well-watered plants [48]. Similar effects were reported in *B. carinata* cultivar 'Merawi', where a 15-day drought period led to a 15% decrease in leaf number and an 86% reduction in leaf area compared to control plants [49]. These morphological changes represent adaptive strategies to limit water loss through transpiration, which is consistent with earlier findings that drought stressed plants reduce their leaf area and number to minimize water loss [50,51]. In the current study, populations E and F showed decreased productivity under 50% irrigation due to reduced leaf area, while in population C, both the number of leaves and leaf area were significantly decreased. However, a statistically significant increase in dry matter content was observed in populations C and E under drought conditions. Notably, some rapeseed genotypes demonstrated superior drought tolerance due to anatomical and physiological adaptations [52,53]. These findings align with our study, where the populations A, B and D exhibited resilience to a 50% reduction in irrigation, showing no significant differences in biomass, leaf number, or leaf area.

Reduced soil moisture significantly impairs nutrient uptake in plants by limiting nutrient availability through mineralization [54] and restricting their soil mobility via diffusion and mass flow [55]. Nitrate ions are particularly crucial under water deficit conditions, as they help maintain cellular osmotic balance. When nitrates accumulate in vacuoles, they raise intracellular solute concentration, thereby lowering osmotic potential and helping plant cells retain or extract water more efficiently under stress [56]. This role is evident in the findings of Ma et al. [57], where drought-induced nitrate accumulation was observed in *B. napus* and *B. juncea*, contributing by 42–47% to the osmotic adjustment in leaves. As nitrate levels in the edible parts of leafy vegetables are a key quality indicator, the European Union has set maximum permitted limits ranging from 3000 to 5000 mg per kg fresh weight according to Commission Regulation (EC) No. 1258/2011 [58]. In the current study, populations C and E exhibited the highest NO_3 concentrations among the examined

Brassica populations, with population C showing increased NO_3 levels under drought conditions but within the limits established by the EU. This observation is corroborated by the findings of Šola et al. [59] who reported that drought raised nitrate content in *B. oleracea* L. tissues by up to 71%. While nitrate uptake may still occur during drought, the stress often suppresses key metabolic processes, including nitrogen assimilation. This suppression occurs primarily due to reduced nitrate reductase activity, an enzyme vital for converting nitrate into organic nitrogen compounds such as amino acids, thereby leading to further nitrate accumulation [60].

Water deficit significantly affects phosphorus (P) concentrations in *Brassica* leaves. Previous research demonstrates this impact clearly. In the study by Geremew et al. [61], *B. juncea* grown under moderate (50% soil moisture) and severe drought stress (25% soil moisture) showed a reduction in leaf P concentration by 26% and 65%, respectively. Similarly, Biswas et al. [62] documented a 23% decline in leaf P concentration in *B. napus* cv. Nugget following irrigation at 45% field capacity, in contrast to well-watered plants at 85% field capacity. In the present study, a decline in P levels was observed, exclusively in population E, with a relatively modest decrease of 9% under drought conditions. The observed variations across studies could be attributed to differences in drought intensity, the duration of stress, or genetic variability among species and cultivars. The observed decline in P under drought conditions aligns with the hypothesis that, during short-term water stress, plants down-regulate metabolic activity in the shoots to minimize water and nutrient use, while roots become more metabolically active to enhance resource acquisition and mitigate drought effects [63].

Drought stress impairs the absorption and internal transport of nutrients, particularly by limiting K diffusion in the soil toward the root zone, resulting in decreased K uptake [64]. It is imperative to ensure that adequate levels of K are maintained within plants, as this is crucial for effective adaptation to drought stress conditions [65]. The elevated demand for K in the context of abiotic stress has been associated with its involvement in several physiological processes related to stress tolerance, including mechanisms linked to CO_2 assimilation, as reported in previous studies [66]. However, the present study did not directly assess such physiological processes. Previous research by Bahrami-Rad and Hajiboland [67] showed no significant difference in leaf K levels between drought stressed and well-watered tobacco plants. Similarly, in the present study, no notable differences in K concentration were observed between irrigation treatments in *Brassica* leaves. However, a statistically significant 29% increase in Na concentration was detected under drought conditions. According to the study by Mohamadi et al. [68], a significant increase in Na concentration was observed in the leaves of *B. napus* L. cv. Madonna under both mild and severe drought stress conditions. In the present study, significant differences were also found among the *Brassica* populations examined. Population E exhibited the highest K and lowest Na levels, while populations A and B displayed reduced K and increased Na levels. The rise in Na may compensate for diminished K by acting as an osmotic agent within the cell, helping to maintain cation balance [69].

Reduced water availability has been shown to inhibit the uptake and decrease foliar concentration of key micronutrients, such as Fe, Zn and Mn, in *Brassica* species [70]. These micronutrients are essential cofactors for antioxidant enzymes like superoxide dismutase [71] and their reduced availability under drought may potentially affect the activity of such enzymes, although this was not directly assessed in the present study. Fe, Zn, Mn and Cu have been identified as essential cofactors for antioxidant enzymes, which play a crucial role in protecting the human body from diseases associated with oxidative stress caused by free radicals [72]. In the present study, a statistically significant reduction was observed only in Zn and Mn concentrations in the leaves of *Brassica* populations culti-

vated under drought conditions. Similar findings were reported by Geremew et al. [61] in *B. juncea*, where both Zn and Mn levels declined under mild (50% FC) and severe (25% FC) drought stress, whereas Fe decreased significantly only under severe stress. Furthermore, the present study revealed significant variations among populations in the concentration of all micronutrients. Specifically, population D exhibited the lowest concentrations of Fe and Mn, while the commercial variety F demonstrated the lowest Zn levels but the highest Cu content compared with the other populations examined.

In the present work, it was demonstrated that drought stress exerts a significant effect on the biochemical traits of *Brassica* leaves, particularly their antioxidant capacity, total phenolic and flavonoid content. As demonstrated in previous research, a moderate water deficit has been shown to enhance drought tolerance in plants by activating the antioxidant defense system [73]. According to Šola et al. [59], TPC increased by approximately 29% under drought conditions, indicating that the accumulation of phenolic compounds plays a crucial role in the plant's adaptive response to oxidative stress caused by water scarcity. Analogous trends have been documented in diverse genotypes of *B. oleracea* L. [74]. In the present study, statistically significant differences were also observed in all measured parameters FRAP, TEAC, TPC, and TFC across the various *Brassica* populations studied. These findings are consistent with previous reports indicating that both TPC and TFC increase significantly under drought stress [75]. However, only certain phenolic compounds and flavonoids have demonstrated antioxidant activity, so identifying the specific compounds involved would provide clearer insight into the plant's antioxidant response.

This study represents the first systematic evaluation of multiple B. cretica populations for drought tolerance, combining agronomic and biochemical traits to provide novel insights into the adaptive potential of this wild species. While drought stress is well known to reduce growth and productivity in *Brassica* crops, our findings reveal considerable variation among *B. cretica* populations, with some populations showing significant reductions in fresh weight, leaf number, and leaf area, while others maintained stable biomass under reduced irrigation. Moreover, population-specific differences in nutrient dynamics, such as nitrate accumulation and phosphorus reduction, highlight distinct physiological strategies for coping with water deficit. Given that *B. cretica* is the closest wild relative of cultivated *B. oleracea* and often occupies marginal environments, these results underscore its value as a reservoir of adaptive alleles for stress resilience. Although wild *Brassica* have previously been explored mainly for their role in conferring resistance to biotic stressors, this work addresses the under-explored potential of *B. cretica* to contribute to breeding efforts aimed at improving tolerance to abiotic stresses, particularly drought.

5. Conclusions

This study elucidates the multifaceted responses of *B. cretica* populations to drought stress, revealing significant variability in growth, biochemical, and nutritional traits in Mediterranean environments. The capacity of certain populations (A, B and D) to sustain growth, maintain nutrient balance, and activate robust antioxidant defenses under water-limited conditions underscores their potential as genetic resources for cultivating drought-tolerant cultivars. To be precise, drought stress induced significant reductions in biomass, leaf area and leaf number in C and E populations, whereas populations A, B and D exhibited stable growth in the presence of a 50% reduction in irrigation. Water deficit significantly altered leaf mineral composition, increasing NO_3^- and Na^+ concentrations and decreasing P, Zn and Mn. Meanwhile, antioxidant capacity and secondary metabolite accumulation (FRAP, TEAC, TPC and TFC) were consistently enhanced under drought stress. These characteristics, including efficient water usage, nutrient homeostasis, and biochemical resilience, are of particular relevance to agricultural practices in semi-arid

regions experiencing increased drought frequency due to climate change. These findings have the potential to inform targeted selection strategies. The integration of these traits into breeding programs offers a viable pathway to develop *Brassica* varieties that exhibit enhanced resilience and stable productivity under Mediterranean climatic conditions. One notable limitation of this study is its reliance on a controlled soilless system, which may not adequately reflect the complexity of field conditions, where stress levels vary over time and interact with other environmental and biological factors. Future efforts should concentrate on identifying the genetic and physiological drivers of these traits, thus enabling more efficient selection through molecular breeding tools and accelerating the development of climate-resilient crops.

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